

Python: if / else

Programs that ask a question, then **decide what to do**. Type them, run them, then try the little changes.

Two things every program below uses:

- **input("...")** asks the user to type something. Wrap it in **int(...)** to turn typed text into a number.
- **if / elif / else** choose what happens. Don't forget the **colon :** and the **spaces** (indent) on the next line.

How to run: open the file in Thonny or IDLE, press **Run** (F5), and answer in the black window. Each program is saved in your `python` folder.

1. Age Checker `6_age_checker.py`

Asks your name and age, then says something different for big kids.

```
# input() asks; int() makes it a number
name = input("What is your name? ")
age = int(input("How old are you? "))

if age >= 8:
    print("Wow " + name + ", you are a big kid!")
else:
    print("Hi " + name + ", you are a little star!")
```

 **Try this:** add an **elif** age >= 13 that prints "You are a teenager!"

2. Secret Door `7_secret_door.py`

A password checker. `==` asks "are these two the same?"

```
print("=== Secret Door ===")
code = input("Enter the secret word: ")

if code == "dragon":
    print("Correct! The door opens. Welcome in!")
else:
    print("Wrong word. The door stays shut.")
```

 **Try this:** change the secret word, then add an **elif** for a second word like "phoenix".

3. Even or Odd `8_even_or_odd.py` — this is your MATH class, in code!

% gives the remainder. If a number $\div 2$ leaves 0, it is even.

```
num = int(input("Type a whole number: "))

if num % 2 == 0:
    print(str(num) + " is EVEN.")
else:
    print(str(num) + " is ODD.")
```

 **Try this:** test 10, 7, 24 and 51. Then change 2 to 5 to test "counts by 5".

4. Weather Helper `9_weather_helper.py`

elif lets you check **more than two** choices.

```
weather = input("Weather? (sunny / rainy / snowy) ")

if weather == "sunny":
    print("Wear your sunglasses and a hat!")
elif weather == "rainy":
    print("Take an umbrella and your boots.")
elif weather == "snowy":
    print("Put on a warm coat and gloves.")
else:
    print("Hmm, I do not know that weather.")
```

 **Try this:** add an **elif** for "windy". (Careful: "Sunny" is not the same as "sunny"!)

5. Number Duel `10_number_duel.py`

Two numbers go in; **if** / **elif** / **else** find the bigger one.

```
a = int(input("First number: "))
b = int(input("Second number: "))

if a > b:
    print(str(a) + " is bigger.")
elif a < b:
    print(str(b) + " is bigger.")
else:
    print("They are equal!")
```

 **Try this:** also print their total, $a + b$.

6. Mini Quiz `11_mini_quiz.py`

Everything together: input + **if/else** + keeping a score.

```
score = 0

ans1 = input("Q1: What is 7 x 8? ")
if ans1 == "56":
    print("Correct!")
    score = score + 1
else:
    print("The answer is 56.")

print("You scored " + str(score))
```

 **Try this:** add a second question and add 1 to the score when it's right.

Your turn — write these yourself

Challenge 1 — Colour reply. Ask for a colour. Print a different message for **red**, **blue** and **green**, and an **else** for anything else.

Challenge 2 — Rollercoaster. Ask the rider's height in cm. If it is **120 or more**, print "You can ride!" — otherwise "Grow a little more first."

Challenge 3 — Your own idea. Ask the user something, then use **if** / **elif** / **else** to reply.



Build it for real: Secret Door

Take program #2 off the screen and into the real world with your Raspberry Pi + Freenove kit. Do **Step 1** first (just lights & a beep), then add the screen in **Step 2**.



Grown-up helps with the wiring. Always power the Pi **OFF** before connecting wires, and use the 220 Ω resistors with the LEDs.

STEP 1 — lights & buzzer only (no screen yet)

Start here — it's the easiest wiring and there's nothing to install. Type the code and a light + beep react. That's the whole **if** / **else**, now with real lights!

Parts

1 green LED · 1 red LED · two 220 Ω resistors · 1 buzzer · breadboard · jumper wires. **(No screen needed yet.)**

Wiring (BCM pin numbers)

Part	Connect to
Green LED (+ 220 Ω)	GPIO17 (pin 11) → resistor → LED → GND
Red LED (+ 220 Ω)	GPIO27 (pin 13) → resistor → LED → GND
Buzzer	GPIO22 (pin 15) → buzzer → GND

The code `secret_door_simple_pi.py`

```
from gpiozero import LED, Buzzer
from time import sleep

green = LED(17)
red = LED(27)
buzzer = Buzzer(22)

SECRET = "dragon"          # change to your own word!

code = input("Enter the secret word: ")

if code == SECRET:
    print("Correct! Green light on.")
    green.on()
    buzzer.on(); sleep(0.2); buzzer.off()
    sleep(3); green.off()
else:
    print("Wrong word. Red light on.")
    red.on()
    sleep(3); red.off()
```

STEP 2 — add the screen (LCD1602)

Once Step 1 works, add the little LCD screen so the message shows on the door itself. You keep the same lights & buzzer and just add the screen.

What it does

You type a secret word. If it's right, the screen says **"Welcome in"**, the **green** light turns on and the buzzer gives a happy beep. If it's wrong, the screen says **"Door stays shut"** and the **red** light turns on. Same **if / else** — now on a real screen!

Extra part & wiring for the screen

Add the **LCD1602** screen (the I²C one, 4 pins) to the Step 1 build:

Part	Connect to
LCD1602 (I ² C)	GND→GND · VCC→5V · SDA→GPIO2 (pin 3) · SCL→GPIO3 (pin 5)

Get the Pi ready for the screen (grown-up, once)

1. Turn on I²C: `sudo raspi-config` → Interface Options → I2C → Yes.
2. Find the screen's address: `i2cdetect -y 1` (usually **0x27** or **0x3f**).
3. Copy the file `LCD1602.py` from the kit's **"I2C LCD1602"** lesson into the same folder as the program.

The code `secret_door_pi.py`

```
from gpiozero import LED, Buzzer
from time import sleep
import LCD1602                # from your Freenove kit

green = LED(17)
red   = LED(27)
buzzer = Buzzer(22)

SECRET = "dragon"             # change to your own word!

LCD1602.init(0x27, 1)         # use YOUR address if not 0x27
LCD1602.clear()
LCD1602.write(0, 0, "Secret Door")
LCD1602.write(0, 1, "Type the code...")

code = input("Enter the secret word: ")

if code == SECRET:
    LCD1602.clear()
    LCD1602.write(0, 0, "Correct!")
    LCD1602.write(0, 1, "Welcome in :)")
    green.on()
```

```
buzzer.on(); sleep(0.2); buzzer.off()
sleep(3); green.off()
else:
    LCD1602.clear()
    LCD1602.write(0, 0, "Wrong word!")
    LCD1602.write(0, 1, "Door stays shut")
    red.on()
    sleep(3); red.off()

LCD1602.clear()
```

 **Level up once it works:** add an **elif** for a special "boss" code that shows a different message · give 3 tries before it locks · make the buzzer beep twice when correct.

Made for a curious coder. if / else — the superpower that lets a program decide. 